AFC–10397

Sub. Code
1BIT1C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Information Technology

PRINCIPLES OF INFORMATION TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Computer.

2. Why does communication technology needed? Give any two reasons.

3. What do you mean by user interface?

4. State the uses of word processing.

5. What is Information Highway?

6. Write the differences between analog and digital signals.

7. Give any two examples for optical disks.

8. What are the elements of File Management Systems?

9. Define Information System.

10. What is Object Oriented Programming?

Part B (5 × 5 = 25)

Answer **all** the questions choosing either (a) or (b).

11. (a) Explain the six elements of a computer and communication system.

Or

(b) Why are ethics needed for Information Technology? Justify.

12. (a) State and explain the types of application software.

Or

(b) Write a short note on Groupware.

13. (a) State and explicate the practical uses of communications and connectivity.

Or

(b) Elaborate the concepts of virtual office.

14. (a) Illustrate the mechanisms of compression and decompression.

Or

(b) Describe the basic concepts of file management.

15. (a) Give a brief note on MIS.

Or

(b) List and explain the five generations of programming languages.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the latest developments in communication technology.

17. Write the features of any two database softwares.

18. Discuss the different types of communication channels.

19. Give a brief account on Data Management Systems.

20. Explain the six phases of System Analysis and Design.

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AFC–10398

Sub. Code
1BIT2C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Second Semester

Information Technology

PROGRAMMING IN C AND DATA STRUCTURES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are 'C' constants?

2. List out bit-wise operators.

3. Define array.

4. Specify the advantages of pointers.

5. Differentiate structure and union.

6. State the procedures to open and close a file.

7. What is stack?

8. How to represent the linked list?

9. Define tree.

10. State any two applications of binary tree.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the structure of a 'C' program.

Or

(b) Discuss the various types of 'C' keywords and Data types.

12. (a) How to pass an array to a function? Explain with suitable example.

Or

(b) Write a short note on array of pointers.

13. (a) Illustrate the procedures to define the structure within another structure.

Or

(b) How to use the functions fseek() and fwrite()? Give a sample program.

14. (a) Write C program to implement the infix, postfix and prefix using stack.

Or

(b) Write an algorithm to traverse in a Linked list.

15. (a) How do you represent the binary tree? Give a sample program.

Or

(b) Draw a binary tree for the expression :

$A * B - (C + D) * (P / Q)$.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the various looping statements available in C language.

17. How to allocate the memory dynamically? Illustrate with a sample program.

18. Write a C program to store the employee details using Structure.

19. Write a C program to implement the queue operations.

20. How to represent the lists? Give a sample program.

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AFC–10399

Sub. Code
1BIT3C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Information Technology

PROGRAMMING IN C++ AND ALGORITHMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the uses Object oriented Programming.

2. State the uses of reference variable.

3. Define constructor.

4. How does dynamic initialization of objects achieved in C++?

5. What is the purpose of an abstract class?

6. What is polymorphism'?

7. What is topological sorting?

8. Write the conditions for binary search tree.

9. Write a travelling salesman problem.

10. What is a shortest path algorithm?

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Explain control structures in C++ with a program.

Or

(b) Write the uses of inline functions with an example.

12. (a) Discuss Dynamic constructor in detail.

Or

(b) Write a C++ program to implement Two-Dimensional Array.

13. (a) Explicate Operator overloading in C++ with sample program.

Or

(b) Write a C++ program to implement virtual function.

14. (a) Illustrate Depth-First Search algorithm with an example.

Or

(b) Elucidate Breadth-First Search algorithm with an example.

15. (a) What is Coin changing? Explain.

Or

(b) Write about Dijkstra's Algorithm.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the following with examples :
    (a) Array within a class
    (b) Pointer to Member.

17. Explain new and delete operators with examples.

18. What is Inheritance? Explain different types of Inheritances.

19. What is Sorting? Discuss the various types of Sorting.

20. Illustrate the Kruskal's algorithm in detail.

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AFC–10400

Sub. Code
1BIT4C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fourth Semester

Information Technology

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is web browser?

2. Define Java tokens.

3. List out the various relational and logical operators in Java.

4. How the built-in mathematical functions are used in Java programming?

5. How to create a multi dimensional array?

6. Mention the uses of final class.

7. How do you add a class to a package?

8. What is meant by thread synchronization?

9. Mention the life cycle of applet programming.

10. How do you run the applet programs?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the applications of OOP.

Or

(b) Illuminate on 'IF' statement with an example.

12. (a) How are expressions evaluated in Java?

Or

(b) Write a Java program to find smallest of three numbers using If statement.

13. (a) Briefly enumerate on abstract class and visibility control.

Or

(b) Write a Java program to merge two strings into one string.

14. (a) Write the program to creating and accessing a packages in Java.

Or

(b) Confer about thread priority and thread exceptions.

15. (a) Give the general structure of a HTML program.

Or

(b) Write Java program to draw a bus using graphics class.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Elucidate on Java support systems and Java environment.

17. Write a Java program to implement switch statement.

18. Explicate in detail about wrapper class and inheritance in Java.

19. Expound on Exception handling mechanism in Java with example.

20. Design a web page that contains the overall information about your College.

AFC–10401

Sub. Code
1BIT5C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Information Technology

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a relational database?

2. Define entity.

3. What is normalization?

4. What is meant by functional dependency?

5. Define Parallel system.

6. Mention the name of the Network types.

7. Give the syntax for creating table.

8. What is meant by view?

9. What is a Cursor?

10. Define Trigger.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What is an E-R diagram? Explain with an example.

Or

(b) Explain briefly about Database System Applications.

12. (a) What is BCNF? Explain the difference between BCNF and 3-Normal Form.

Or

(b) Explain 1st and 2nd Normal Forms with example.

13. (a) Explain about centralized computer system with neat diagram.

Or

(b) Distinguish homogenous and heterogeneous distributed database.

14. (a) Briefly discuss about sequence with suitable example.

Or

(b) Write short notes on user privileges and roles.

15. (a) How can you create and execute a stored procedure? Explain with an example.

Or

(b) Explain cursor attributes with their syntax.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the architecture of database system with neat diagram.

17. Discuss on multivalued functional dependencies with example.

18. Explain architecture of parallel database with neat diagram.

19. Describe the uses of views.

20. Explain different types of database triggers with suitable example.

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AFC–10402

Sub. Code
1BIT5C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Information Technology

VISUAL PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are Subroutines?

2. Differentiate the variables and constants in VB.

3. What are window forms?

4. List out the properties of textbox control in VB.

5. Write the uses of common dialog controls.

6. Define MDI.

7. Expand ODBC.

8. How can you create data reports in VB?

9. List out any two of CLIST resources.

10. Write any two fundamental uses of MFC.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a VB program to perform basic arithmetic calculator.

Or

(b) What do you mean by arguments? How can you passing arguments in calling procedures?

12. (a) What are the control methods available in List box control?

Or

(b) Write note on text box control with an example.

13. (a) Illustrate with an example the text formatting properties in MDI.

Or

(b) Write note on Rich Text Box Control methods.

14. (a) Write short note on data environment and data reports in VB.

Or

(b) Discuss about Active Data Object.

15. (a) Discuss about the fundamentals of MFC.

Or

(b) Write note on Bitmaps and versions.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the various control statements in VB.

17. Write a VB program to calculate the age based on the date of birth.

18. Write brief note on Rich Text Box Control methods with an example.

19. Describe about ActiveX Data Object and Data Connection with an example.

20. Discuss the following in VC++ :

    (a) CWinApp

    (b) CDdiobject and CDialog.

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AFC–10403

Sub. Code
1BITE1B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Information Technology

Elective — MULTIMEDIA TECHNOLOGY AND ITS APPLICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Multimedia?

2. List out any two Multimedia Products.

3. Define Hardware.

4. What is Hypertext?

5. List out any Two Elements of Graphics.

6. Expand MIDI.

7. What is Storyboard?

8. Define Computer Animation.

9. Write short notes on Web Authoring.

10. Expand HTML.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Briefly discuss about Copyright.

Or

(b) Write short notes on Resources of Multimedia Developers.

12. (a) Illustrate Elements of Text.

Or

(b) Write notes on Text in Multimedia Applications.

13. (a) List out characteristics of Sound and Digital Audio.

Or

(b) Write notes on :

(i) Images in Multimedia

(ii) Graphics File Formats.

14. (a) Explain characteristics about Digital Video.

Or

(b) Explain Content Organizational Strategies.

15. (a) Briefly explain Categories of Authoring Tools.

Or

(b) Discuss about strategy for selecting a tool.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss History of Multimedia.

17. Explain Multimedia Computer Architecture.

18. Illustrate MIDI and its related applications.

19. Specify Digital Video in Multimedia Applications.

20. Explain about Multimedia considerations for the Internet.

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AFC–10404

Sub. Code
1BITE2A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Information Technology

Elective — COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define computer networks.

2. What is LAN?

3. Expand SLIP.

4. What is fragmentation?

5. What are the uses of Firewalls?

6. Expand RARP.

7. Define multiplexing.

8. Write any two uses of TCP.

9. Define Multimedia.

10. Expand DNS.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a short note on Telephone Systems.

Or

(b) Expound in short on Communication Satellites.

12. (a) Write short note on HDLC.

Or

(b) Explicate briefly on collision free protocols.

13. (a) Give a note on ATM LANs.

Or

(b) Discuss about Fragmentation.

14. (a) Briefly explain on Crash Recovery.

Or

(b) Illustrate the concept of buffering.

15. (a) Explicate on cryptography.

Or

(b) Write short note on SNMP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the different layers in OSI reference model.

17. Elaborate the concept of Dlink protocols.

18. Describe any four routing algorithms.

19. Briefly explain the concept of Internet Transport Protocols.

20. Write brief note for the following

(a) Electronic Mail

(b) Electronic Mail Privacy.

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AFC–10405

Sub. Code
1BIT6C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Information Technology

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Software Engineering.

2. Differentiate Validation and Verification.

3. What are the software cost factors?

4. What is SDLC?

5. List the fundamental activities of a software process.

6. Write the merits of incremental model.

7. List the task regions in the Spiral model.

8. State the uses of testing.

9. What do you mean by Predictable Risk?

10. What is SQA?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the different phases involved in Waterfall Life Cycle.

Or

(b) Illustrate Spiral model and win-win spiral model in detail.

12. (a) Discuss the prototyping approaches in software process.

Or

(b) Elucidate the concepts of Structural Modeling.

13. (a) Describe the design principles in detail.

Or

(b) What is SCM? Discuss it.

14. (a) What are the advantages and disadvantages of big-bang? Explain them.

Or

(b) Explicate the procedures of Black box testing.

15. (a) List and explain the activities of SQA.

Or

(b) Discuss on ISO 9000 quality standards.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Draw and Explain about the Phase Life Cycle Model.

17. Illustrate any two software cost estimation techniques.

18. Describe the design steps of the transform mapping.

19. Elucidate any two testing strategies.

20. Discuss on formal technical reviews.

AFC–10406

Sub. Code
1BIT6C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Information Technology

MOBILE COMMUNICATION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Antennas?
2. Define Modulation.
3. Expand: GSM.
4. What is Localization?
5. What is IEEE?
6. Expand ATM.
7. What is Mobile IP?
8. Explain snooping.
9. Describe WWW.
10. Expand HTML.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is Wireless transmission? Explain in detail.

Or

(b) Write short notes on cellular systems.

12. (a) Clarify the concept of DECT.

Or

(b) Write the a short note on Cyclic Representation of Data.

13. (a) Explain various features available in WLAN.

Or

(b) What is Location Management? Explain in detail.

14. (a) Elucidate the concept of Mobile network layer.

Or

(b) How to use Transaction oriented TCP?

15. (a) What are the features of using HTTP?

Or

(b) Explain the WAP Architecture.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the concept of Spread Spectrum.

17. Explain the following

    (a) SDMA and CDNMA

    (b) Digital Audio Broadcasting

18. Clarify the concept of WATM Services.

19. Describe the concept of Dynamic host configuration protocol.

20. Explain the System Architecture

AFC–10407

Sub. Code
1BIT6C3

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Information Technology

WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Expand HTML
2. What does <b> tag do?
3. Write the Syntax for 'if' structure in Java Script.
4. List any two logical operators in Java Script.
5. What is an array?
6. Give an example for a String and Date object.
7. Why are events used?
8. Mention any two Advanced filters in DHTML.
9. What is scaling?
10. How shapes are moved?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Design a web page in HTML that Utilizes common tags such as paragraph, bold and horizontal rules.

Or

(b) Create a list of friends and places to visit using ordered and unordered lists.

12. (a) Write a Javascript program to find if a number is odd or even.

Or

(b) Illustrate the do/while structure in Javascript with an example.

13. (a) Mention the procedure of declaring and allocating arrays with an example.

Or

(b) Briefly explain about Boolean and Number objects.

14. (a) Give a short note on Dynamic styles positioning.

Or

(b) Elucidate the IMG element with suitable examples.

15. (a) How do you bind data to an IMG and Table? Explain with steps.

Or

(b) Discuss about Mouse events and External source files.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Create a webpage in html for a children's book publishing company, using all common tags. Give suitable illustrations.

17. List and explain the different types of operators available in JavaScript with examples.

18. Give notes and suitable examples on the following with respect to JavaScript :

    (a) Passing arrays to functions

    (b) Recursion.

19. List an explain any 5 filters in DHTML with suitable illustrations and examples.

20. Explain in detail about graphic manipulation and Path control animation in Dynamic HTML.

AFC–10408

Sub. Code
1BITE3A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Information Technology

***Elective* – DISTRIBUTED COMPUTING**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Mention two uses of Web Search.

2. Expand MMOG.

3. What is an API?

4. Enumerate any two connection protocols.

5. Define Middleware.

6. What is a thread?

7. Expand DNS.

8. Define External Synchronization.

9. What is a Nested Transaction?

10. Define Timestamp.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Give a note on Resource Sharing.

Or

(b) List and explain the types of networks.

12. (a) Discuss in brief about Remote Method Invocation.

Or

(b) Mention the methodology behind Multicast Communication.

13. (a) What is meant by Virtualization at Operating System Level? Explain.

Or

(b) Give a short note on Symmetric and Asymmetric cryptographic algorithms.

14. (a) Discuss about the functionality of Clock skew and Clock drift.

Or

(b) Explicate the necessity for Directory Services.

15. (a) What is meant by Atomic commit protocols? Clarify.

Or

(b) Elucidate about Passive Replication.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Enumerate the challenges encountered by Distributed Systems and explain them.

17. Describe about Remote procedure Call in detail.

18. Discuss in detail about the Operating System Architecture with illustrations.

19. Elucidate the Global States in Time Control.

20. Illustrate the importance of handling Distributed Deadlocks in detail.

AFC–10409

Sub. Code
1BITE3B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Information Technology

Elective — DIGITAL IMAGE PROCESSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is image processing?
2. List down elements of digital image processing.
3. What is Fourier transform?
4. List out any two features of image transform.
5. What is filtering?
6. Explain the Least square restoration.
7. What is thresholding?
8. What is image compression?
9. Define Interpretation.
10. What is Morphology?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the uses of photographic film.

Or

(b) Discuss the various steps in Image Processing.

12. (a) Write a short note on hotelling transform.

Or

(b) Explain the properties of the two dimensional fourier transform.

13. (a) Narrate Spatial filtering.

Or

(b) Define Inverse filtering and its advantages.

14. (a) Explain the image compression model.

Or

(b) Describe about Lossy Compression.

15. (a) Explain the Boundary Descriptors in representation schemes.

Or

(b) Discuss the Relational Descriptors.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a detailed note on the elements of visual perception.

17. Discuss the fast Fourier transform in detail.

18. Explain Image Restoration and color image processing.

19. Discuss about the elements of information theory.

20. Explain the various elements of image analysis.

AFC–10504

Sub. Code
1BITSA1/ 1BIT1A1

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Information Technology

Allied: DISCRETE MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is compound statement? And give an example.

2. Show that the statement P^7P is a contradiction.

3. Define Conjunctive Normal Form of a statement.

4. Write the inference rules of hypothetical syllogism and disjunction syllogism.

5. What is simple graph? And give an example.

6. Define circuit with an example.

7. What is meant by minimum spanning tree?

8. Define cut vertices with an example.

9. Define binary relation from one set to another and give an example.

10. What is partial ordering? And give an example.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Construct the truth table for $(P \vee Q) \vee 7P$.

Or

(b) Define conditional and biconditional statement.

12. (a) Determine whether the conclusion C follows logically from the premises H_1 and H_2.

(i) $H_1 : P \to Q \quad H_2 : 7(P \wedge Q) \quad C : 7P$

(ii) $H_1 : 7P \quad H_2 : P \leftrightarrow Q \quad C : 7(P \wedge Q)$

Or

(b) Show that RVS follows logically from the premises CVD, $(CVD) \to 7H, 7H \to (A \wedge 7B)$ and $(A \wedge 7B) \to (RVS)$.

13. (a) Explain degree of a vertex with example.

Or

(b) Explain isomorphism of two graphs with an example.

14. (a) Write Prim's Algorithm to find the minimum spanning tree of a graph.

Or

(b) Explain Hamiltonian graph with suitable example.

15. (a) Draw the Hasse diagram for $\leq$ relation on *{0, 2,5,10,11, 15}*.

Or

(b) State and prove isotonic property of lattices.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the following term:

    (a) Negation (b) Conjunction

17. Obtain the principal Conjunctive Normal Form of *(7(PVQ)) V (P^Q).*

18. Explain bipartite graph with suitable example.

19. Explain Dijkstra's Algorithm to find the shortest path between a specified vertex to another specified vertex in a graph.

20. Write the properties of Boolean Algebra.

AFC–10505

Sub. Code
1BITSA2

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Information Technology

***Allied* — OPERATION RESEARCH**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Operation Research?

2. Define a model.

3. What is slack variable?

4. Define Optimum Solution.

5. What is integer linear programming?

6. What are the advantages of the branch and bound method?

7. Write the mathematical formulation of an assignment problem.

8. What is travelling salesman problem?

9. What do you mean by transportation model?

10. Define balanced transportation problem.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the applications of OR.

Or

(b) Write a short note on the importance of operation research in production management.

12. (a) Write the procedure for forming a LPP Model.

Or

(b) Solve the following L.P.P graphically

Maximize $Z = 100X_1 + 40X_2$

Subject to $5X_1 + 2X_2 \leq 1000$

$3X_1 + X_2 \leq 900$

$X_1 + 2X_2 \leq 500$ and

$X_1, X_2 \geq 0.$

13. (a) Explain the relationship of the dual and primal problems.

Or

(b) Use dual simplex method to solve the L.P.P.

Max $Z = 2X_1 + 3X_2$

Subject to $2X_1 - X_2 - X_3 \geq 3$

$X_1 - X_2 + X_3 \geq 2$

$X_1, X_2, X_3 \geq 0$

14. (a) What is the difference between assignment problem and travelling salesman problem?

Or

(b) Solve the assignment problem

	A	B	C	D
I	1	4	6	3
II	9	7	10	9
III	4	5	11	7
IV	8	7	8	5

15. (a) Find the initial basic feasible solution for the following transportation problem by least cost method

To

		A	B	C	D	Supply
	P	1	2	1	4	30
From	Q	3	3	2	1	50
	R	4	2	5	9	20
	Demand	20	40	30	10	

Or

(b) Solve the transportation problem

Destination

		A	B	C	D	Supply
	1	11	20	7	8	50
Source	2	21	16	20	12	40
	3	8	12	18	9	70
	Demand	30	25	35	40	

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the steps involved in the solution of an operation research problem.

17. Use Big-M Method to solve

Minimize $Z = 4X_1 + 3X_2$

Subject to $2X_1 + X_2 \geq 10$

$-3X_1 + 2X_2 \leq 6$

$X_1 + X_2 \geq 6$ and

$X_1 + X_2 \geq 0.$

18. Solve the following integer programming problem using the branch and bound method

Maximize $Z = 3X_1 + 5X_2$

Subject to the constraints

$2X_1 + 4X_2 \leq 25$

$X_1 \leq 8$

$2X_1 \leq 10$

$X_1, X_2 \geq 0$ and integers.

19. Solve the following travelling salesman problem

		To			
		A	B	C	D
	A	–	46	16	40
From	B	41	–	50	40
	C	82	32	–	60
	D	40	40	36	–

20. Find the initial basic feasible solution for the following transportation problem by VAM.

		Distribution centre				
		D1	D2	D3	D4	Availability
	S1	11	13	17	14	250
Origin	S2	16	18	14	10	300
	S3	21	24	13	10	400
	Requirements	200	225	275	250	

AFC–10506

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1BITSA3

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Information Technology

Allied — ACCOUNTING PRINCIPLES AND COMPUTER APPLICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

Answer in 1 or 2 sentences.

1. State the limitations of recording transactions in the Journal.

2. What do you mean by Practical system or English system of book keeping?

3. Classify the types of errors.

4. Define the term Ledger.

5. What is a debit note?

6. Write any two advantages of Petty Cash Book.

7. What is a purchase returns book?

8. What is bills receivable book?

9. Define Bank Reconciliation.

10. What is meant by Double Cash Entry?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Explain the different types of accounting and objectives of accounting.

Or

(b) Define the following :

(i) Transaction

(ii) Journal

(iii) Ledger

(iv) Debtor.

12. (a) What is a Profit and Loss Account? Write its format neatly.

Or

(b) Write journal to rectify the errors in the following entries.

(i) Cash Sales for Rs.30,000 was completely omitted.

(ii) Purchase from Arun and Co cash Rs.400 was not credited to cash a/c. However purchase a/c was debited correctly.

(iii) Sales book is overcast by Rs.500

(iv) Purchase book is undercast by Rs.1,000.

13. (a) Enter the following transactions in the purchase book of Arun Kumar :

May 1 Bought from Mr. 'A' 10 boxes of Nirma soap @ Rs 150 per box @ discount of 20%

8 Purchased from Mr. 'B' 5 boxes of Vim detergent powder Rs 60 per box, for cash.

10 Received from Mr. 'C' 120 pieces of sandalwood soap @ Rs. 8 per piece, less discount 20%

15 Purchased furniture from Mr. 'D' Rs. 5,548.

18 Received invoice from Mr. 'E' for 15 cases of "Wash bar" @ Rs. 180 per box.

12 Cases of washing powder Rs. 60 per cake. Forwarding charges Rs. 20 less 25% discount

30 Purchased office stationery from Mr. 'A' Rs. 180.

Or

(b) Prepare a petty cash book on the Impress system from the following :

2005	Rs.
March 1	Received Rs. 1000 for petty cash
March 2	Paid bus fare 10
March 3	Paid Cartage 25
March 4	Paid for postage and telegram 50
March 5	Paid for stationery 40
March 6	Paid for postage and telegram 50
March 7	Paid for sundry exp. 40

14. (a) Prepare Profit and loss a/c for the following Trial Balance of Narayana Rao & Co. on 3 1.12.2007 :

Account	Debit Rs.	Credit Rs.
Purchases	1,00,000	
Wages	6,000	
Rent	2,400	
Travelling expenses	4,800	
Interest	1,200	
Returns inward	4,000	
Bank	10,000	
Cash	34,000	
Machinery	14,000	
Furniture	1,000	

Account	Debit Rs.	Credit Rs.
Loan		45,800
Miscellaneous expenses	200	
Returns outward		3,000
Salaries	12,000	
Insurance	800	
Discount	900	
Sales		99,900
Sundry creditors		50,000
Capital		1,10,000
Drawings	15,000	
Advertisements	2,400	
Buildings	10,000	
Sundry debtors	80,000	
Stock (1.1.2007)	10,000	
	3,08,700	3,08,700

Or

(b) Explain the difference between Profit and Loss Account and Balance Sheet.

15. (a) Enter the following transactions in a simple cash Book.

2006		Rs.
Jan 1	Cash in hand	12,000
Jan 5	Received from Ram	3,000
Jan 7	Paid Rent	300
Jan 8	Sold goods	7,000
Jan 10	Paid Shyam	2,000

Or

(b) Write and explain the format for Bank Reconciliation.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Journalize the following transactions and prepare a trial balance: Rex commenced business with a capital of Rs.1,00,000 purchased furniture for Rs.5,000. Paid Rent for Rs.2,000. Sold goods for Rs.25,000. Purchased for Rs.20,000. Sold goods to Thomas on credit Rs.10,000. Paid Electricity charges Rs.3,000.

17. The following are the balances extracted from the books of Mr. Mathew as on 31.3.2011.

	Rs.		Rs.
Capital	15,000	Loans borrowed	20,000
Drawings	4,800	Stock (1.4.2008)	7,500
Machinery	20,000	Purchases	60,000
Furniture	1,500	Sales	90,000
Sundry Debtors	20,000	Office rent	1,100
Sundry Creditors	13,000	Insurance	240
Interest (dr)	1,250	Discount allowed	1,000
Wages	10,000	Discount earned	500
Salaries	7,500	General expenses	1,200
Carriage inwards	500	Cash on hand	150
Purchase returns	1,000	Bank balance	1,260
Sales returns	1,500		

Adjustments :

(a) Stock on 31.3.2011 10,000

(b) Rent outstanding 100

(c) Salaries outstanding 900

(d) Insurance prepaid 40

(e) Interest on loans outstanding 250

(f) Provide depreciation 10% on Machinery, 6% on furniture

Pass necessary adjusting entries and prepare trading and profit and Loss account for the year ended 31.3.2011 and a balance sheet as at the closing date.

18. Write a detailed account on Subsidiary Books.

19. (a) Prepare a Sales Return Book from the following particulars :

2008

Feb 1 : Sold to Rishi & Co., Bangalore, 10 bags of rice at Rs. 220 per bag.

Feb 7 : Sold to Kumar Chennai, 10 bags of sugar at Rs. 350 per bag

Feb 12 : Sold to Sashi & Sashi Co., Bombay. 10 bags of wheat at Rs. 70 per bag

Feb 18 : Sold to Rakesh & Co., Kolkata, 10 bags of groundnut at Rs. 200 per bag

Feb 25 : Sold to Chand & Co., Bangalore, 10 bags of wheat flour at Rs. 150 per hag

(b) Prepare a Purchase Return Book from the following particulars :

2008

Oct 1 : Returned goods worth Rs. 300 to Rahul and Co., Chennai

Oct 9 : Returned goods worth Rs. 100 to Raju. Kolkata

Oct 15 : Allowance claimed from Ashok and Co., for shortage Rs. 150

20. Rocky's Cash Book (Bank Column only) for the month of June 2008 was as follows :

Dr.		($)	Cr.		($)
June 1	Balance b/f	600	June 3	Tim	90
June 14	Cash	225	June 19	Sam	270
June 26	Roby	60	June 23	Rakesh	1050
June 30	Balance hint 1	525			
		1410			1410

The following Bank Statement was received in early July 2008 :

2008	Details	Payments	Receipts	Balance
June 1	Balance brought forward			520
1	Error corrected-contra 31 May	hint 2	80	600
14	Cash			825
15	Tim	90		735
20	Banker's Order-fire insurance	85		650
21	Sam	270		380
26	Roby		60	440
29	Unpaid cheque – Roby	60		380
30	Dividend		90	hint 3 470

You may assume that the bank balance was successfully reconciled at the end of March.

(a) Bring the Cash Book up to date, starting with the balance at 30 June 2008.

(b) Prepare a statement, under its correct title, to reconcile the difference between your amended Cash Book balance and the balance in the Bank Statement on 30 June 2008.

(c) State the amount of the up-dated bank balance. hint 4

Hint 1 : This is a credit balance i.e. an overdraft

Hint 2 : This entry has already appeared in Cash Book in May. Note the date 31 May. Note also the difference between opening balance in Cash Book ($600) and opening balance in Bank Statement ($520) is $80. So no adjustments in June.

Hint 3 : This is a credit balance. A credit balance in Bank Statement is equivalent to a debit balance in Cash Book.

Hint 4 : Updated means, alter all necessary adjustments have been made.

AFC–10507

Sub. Code
1BITSA4

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Information Technology

Allied — E-Commerce

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Internet Commerce.

2. What do you mean by Internet Society?

3. Define Web.

4. Specify the features of Smart Cards.

5. Name any two applications of EDI.

6. What is EDI envelope?

7. State any two reasons to advertise on Internet.

8. Define Data Interface.

9. State any advantages of Computer-based Training.

10. Write any two properties of agents.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the Components of I-Way.

Or

(b) Describe the evolution of Internet.

12. (a) What are the security mechanisms adopted for the Web? Illustrate them.

Or

(b) Illustrate the concepts of Digital Token-based EPS.

13. (a) Elaborate the legal issues of EDI.

Or

(b) How to implement the EDI software? Explain its procedures.

14. (a) Explain the different processes of Online Marketing.

Or

(b) How to filter the information? Discuss its mechanisms.

15. (a) Explicate the technological components of Computer-based Education.

Or

(b) Describe the characteristics of software agents.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly explain the anatomy of E-Commerce applications.

17. List and explain the risks of Electronic Payment Systems.

18. Give a brief account on Value Added Networks on EDI.

19. Discuss the new age of Information-based Marketing.

20. Give a brief account on Digital Copyrights.